

RENSSELAER MOTORSPORT

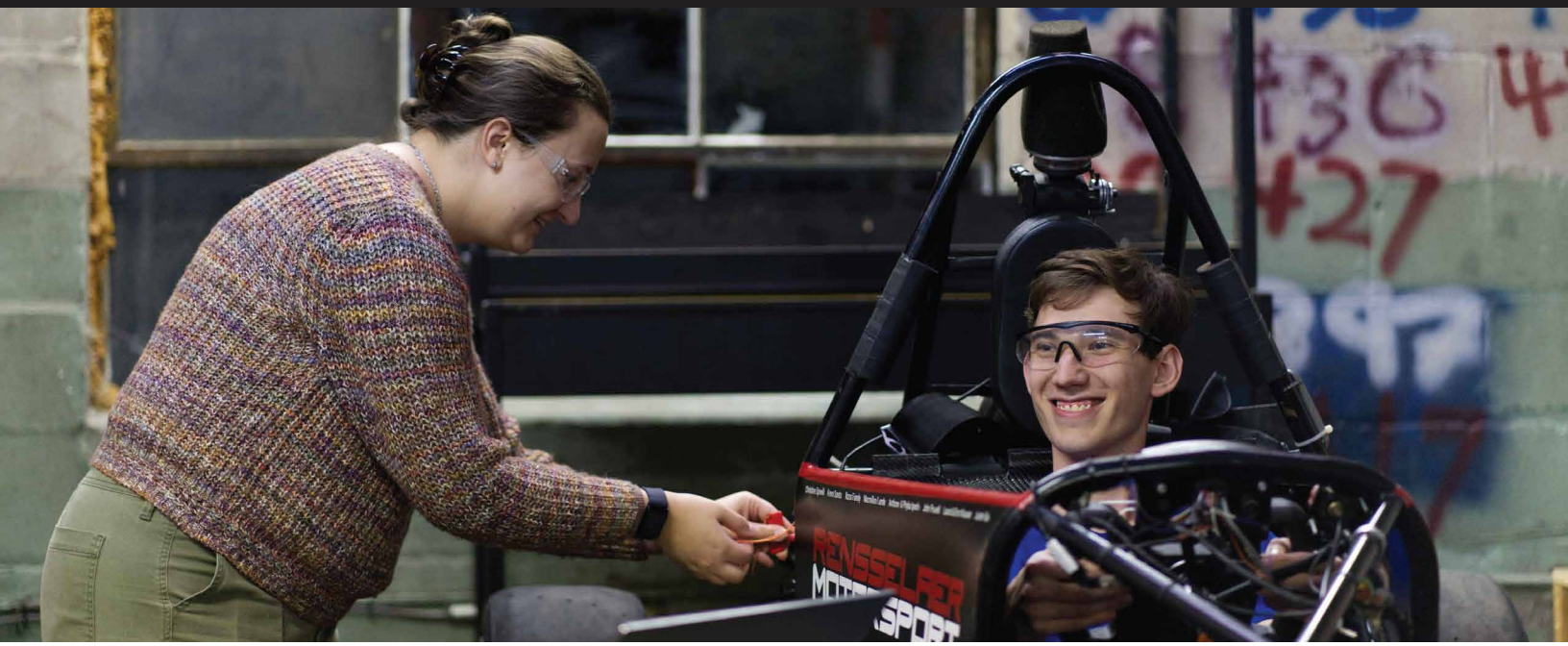


RM28

Sponsorship Packet

2024-2025

Rensselaer Motorsport, EST. 1992, rensselaermotorsport.com, @rensselaer_motorsport
RPI Formula SAE, MANE JEC 5049, 110 8th St., Troy NY 12180



We are Rensselaer Motorsport, Rensselaer Polytechnic Institute's Formula SAE team. We are a group of students who are devoted to engineering excellence. Since attending our first competition in 1992, we've grown substantially. Every single year we design, prototype, manufacuture, validate, and test our formula-style open-wheel car. We do so by tirelessly using interdisciplinary problem solving, engineering design, science, and management prectices to build the most reliable, efficient, and fast race car.

Where We've Gone

Honda Performance Development
Stewart-Haas Racing
Tesla Motors
Multimatic
Lockheed Martin
General Dynamics
Markforged
General Motors
Sensata Technologies
General Electric
SpaceX
Boeing

Our Majors

Mechanical Engineering
Aeronautical Engineering
Computer Systems Engineering
Electrical Engineering
Computer Science
Design, Innovatoin, and Society
Industrial and Management Engineering
Sustainability Studies
Chemistry

1992-2006 2006-2013 2013-Present

Team Leadership



Team Lead
Zayd Farjo
Mech Eng. '26



Technical Director
Erich Bahm
Mech Eng. '25



Shop Manager
Nicholas Cummings
Mech Eng. '27

Business Team



Business Lead
Joey Marchand
Mech Eng. '27

Vehicle Dynamics (VD) Team



Vehicle Dynamics Lead
Joe Richter
Aero Eng. '25



Chassis/Suspension/ Tires Lead
Grace Yu
Mech/Aero Eng. '26



Aerodynamics Lead
Theodore Racz
Mech/Aero Eng. '27



Cooling Lead
Elliot Wilk
Mech/Aero Eng. '26



Brakes Lead
Nick Lagaros
Computer Sys Eng. '28

Electric Vehicle (EV) Team



EV and HV Lead
Sean Trimper
Electrical Eng. '25



LV Design Lead
Blake Hopper
Electrical Eng. '25



LV Mfg. Lead
Lauren Bauer
Chemistry '25



Drivetrain Lead
Joey Marchand
Mech Eng. '27



Controls Lead
Amar Maksumic
CS/CSE '25



Accumulator Lead
Maya Gorse
Mech Eng./DIS '27

RM28 will be an electric vehicle!

EMRAX 228 Combined-Cooled Electric Motor

4130 Steel Tube Frame Chassis with Custom Aerodynamics

ISR & Wilwood Brake Calipers with Custom Brake Rotors

Custom Designed Battery Pack (Accumulator)

RM28 will be a new and improved racecar! The key difference from last year is the addition of prepackaged power modules that we assemble into segments, and then a full battery pack. The switch to the modules will allow us to run higher peak power, and focus more time on testing/ the iterative design process.

SAE INTERNATIONAL®

The Formula SAE (Society of Automotive Engineers) competition is the largest intercollegiate design competition in the world. Rensselaer first competed in Formula SAE in 1992, and over the last 29 years we've celebrated top 10 finishes, forged new and innovative designs, and provided hundreds of students with the experience of a lifetime.













Formula SAE Michigan Competition Structure

We design a new car each year adhering to a set of rules determined by Formula SAE. Our car is designed to pass a rigorous electric vehicle technical inspection, as well as perform in multiple static and dynamic events. These events are described below.

Technical Inspection

Accumulator Inspection:

A rigorous inspection of a teams' battery pack ensuring compliance and safety.

Static Tech:

Overall rules inspection for car compliance and safety.

Rain Tech:

An EV specific inspection, in which the car is doused with water to ensure waterproofing.

Tilt Test:

The car is tilted to 45 and 60 degrees to ensure there are no fluid leaks and the car will not flip while driving.

Brake Tech:

A test to ensure all the brakes are functional and can lock the wheels at 30 mph.

Static Events (325 Points)

Business Presentation (75 Pts):

We are asked to present a business model for our team in response to a prompt each year, similar to: "How have the recent changes in the supply chain affected you and how have you responded?"

Cost Event (100 Pts):

Each team presents a full bill of materials of their vehicle and gets a score relative to other teams' build price. Teams also get scored on a real-case scenario.

Design Event (150 Pts):

The design event is the heart of the FSAE competition: students are asked to defend every single portion of their car's design by showing their engineering design process and the validation done throughout the season.

Dynamic Events (675 Points)

Skidpad (75 Pts):

A figure-8 pattern to measure vehicle cornering ability around a constant radius turn.

Acceleration (100 Pts):

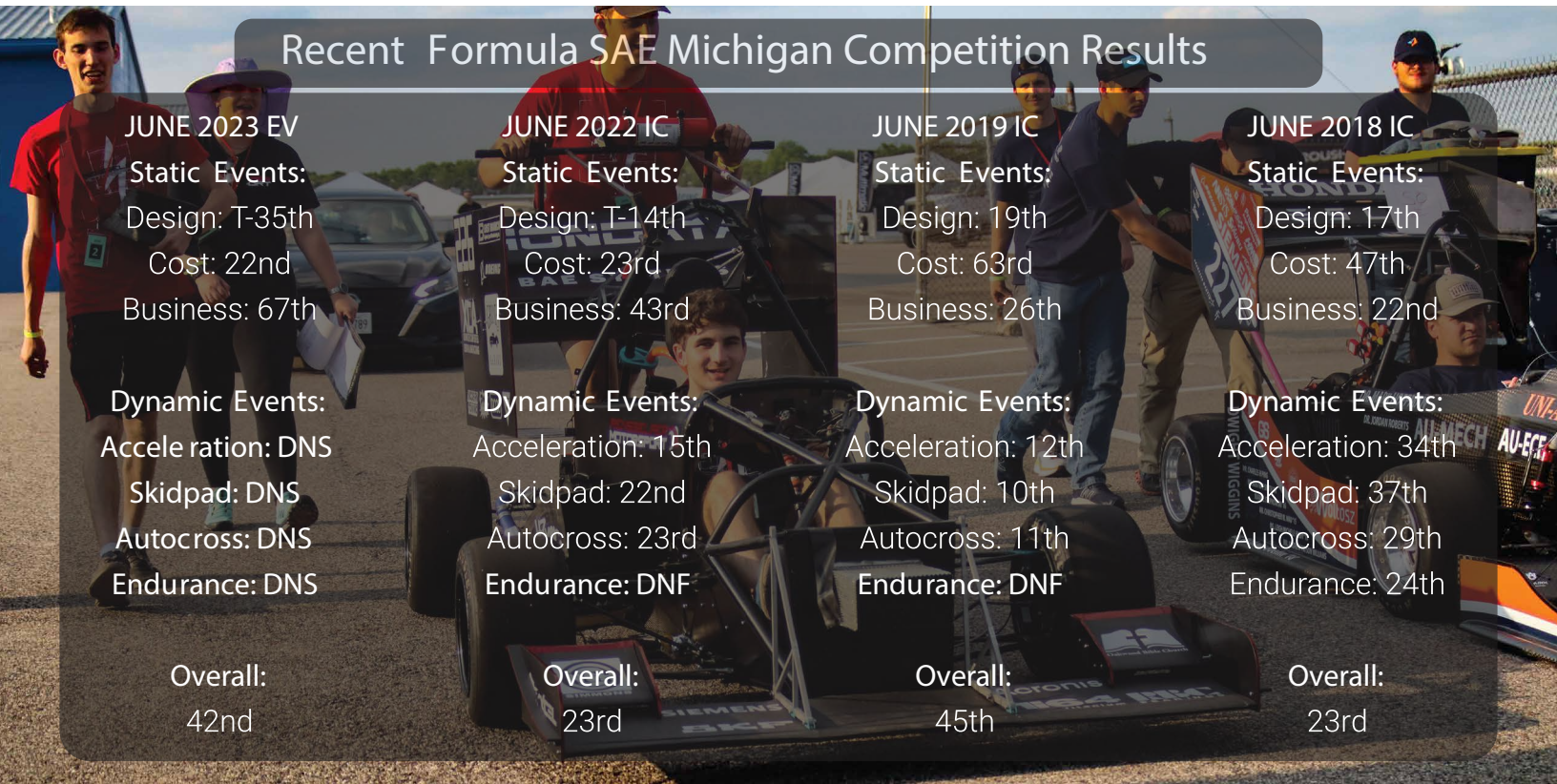
75 meter drag run to evaluate vehicle acceleration in a straight line on flat pavement.

Autocross (125 Pts):

Cone course autocross event designed to evaluate vehicle maneuverability and handling through a tight course.

Endurance & Efficiency (275 & 100 Pts):

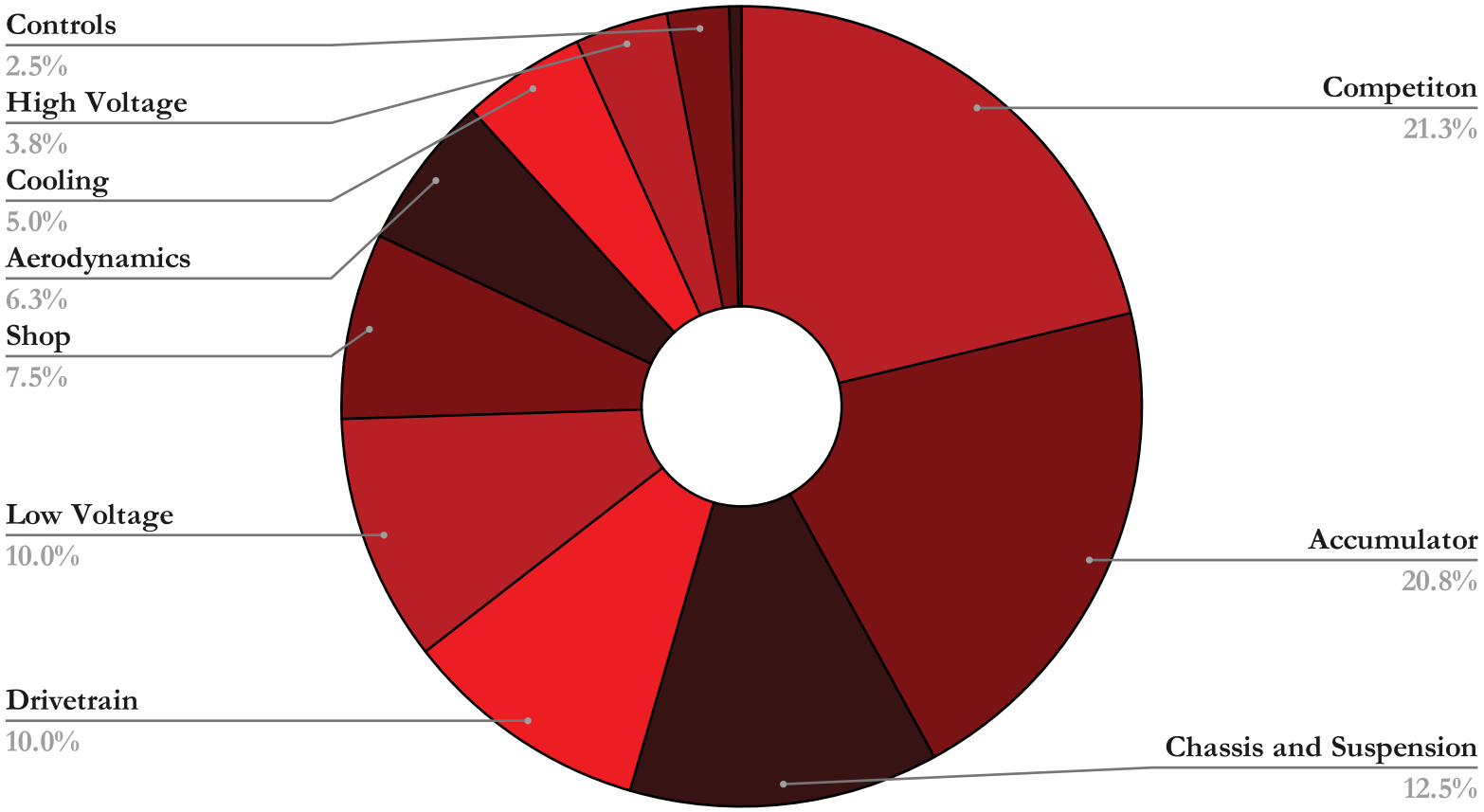
22km constant race with a driver change half way through to test the durability and reliabity of the vehicle, along with measuring energy use throughout to determine efficiency.



Recent Formula SAE Michigan Competition Results

JUNE 2023 EV	JUNE 2022 IC	JUNE 2019 IC	JUNE 2018 IC
Static Events:	Static Events:	Static Events:	Static Events:
Design: T-35th	Design: T-14th	Design: 19th	Design: 17th
Cost: 22nd	Cost: 23rd	Cost: 63rd	Cost: 47th
Business: 67th	Business: 43rd	Business: 26th	Business: 22nd
Dynamic Events:	Dynamic Events:	Dynamic Events:	Dynamic Events:
Accele ration: DNS	Acceleration: 15th	Acceleration: 12th	Acceleration: 34th
Skidpad: DNS	Skidpad: 22nd	Skidpad: 10th	Skidpad: 37th
Autocross: DNS	Autocross: 23rd	Autocross: 11th	Autocross: 29th
Endurance: DNS	Endurance: DNF	Endurance: DNF	Endurance: 24th
Overall: 42nd	Overall: 23rd	Overall: 45th	Overall: 23rd

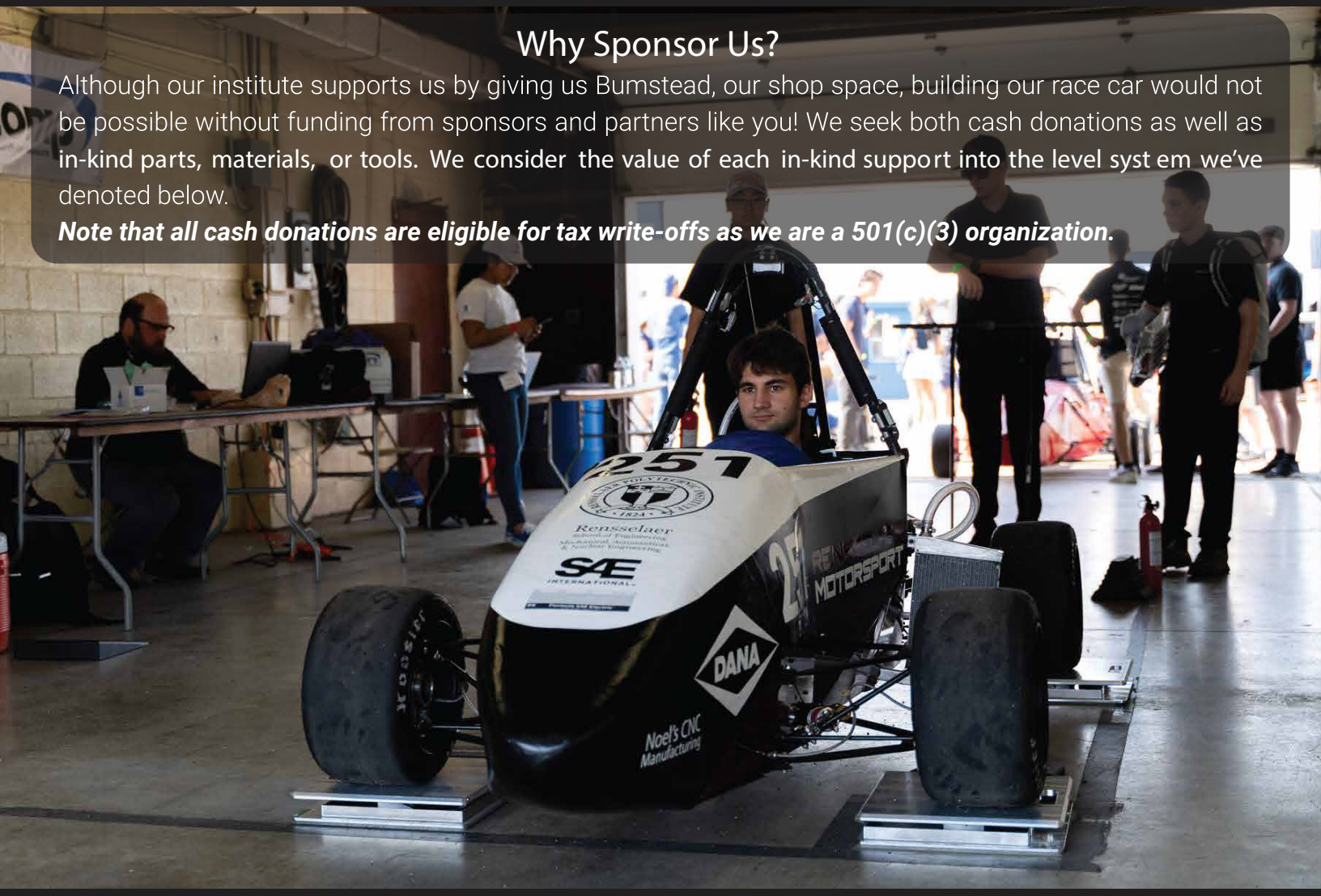
RM28 Projected Budget



This year, we need your support more than ever! The transition to EV is still actively happening, and we're so close! Here is our budget breakdown by subsystem, with our biggest expenses shown in red:

Competition	$17,000.00	High Voltage	$3,000.00
Rollout	$400.00	Low Voltage	$8,000.00
Shop	$6,000.00	Controls	$2,000.00
Chassis/Suspension	$10,000.00	Accumulator	$16,600.00
Aerodynamics	$5,000.00	Cooling	$4,000.00
Drivetrain	$8,000.00		

Grand Total Projection: $80,000



Why Sponsor Us?

Although our institute supports us by giving us Bumstead, our shop space, building our race car would not be possible without funding from sponsors and partners like you! We seek both cash donations as well as in-kind parts, materials, or tools. We consider the value of each in-kind support into the level system we've denoted below.

Note that all cash donations are eligible for tax write-offs as we are a 501(c)(3) organization.

Sponsor Tiers

	Title Sponsor $8,500+	Level 3 $5,000+	Level 2 $2,500+	Level 1 $1,000+
Access to Monthly Newsletter	✓	✓	✓	✓
Logo on our Website	✓	✓	✓	✓
Logo on Car	Large	Medium	Small	Name Only
Invitations to Events	All Events	All Events	Rollout Only	
Logo on Team Merchandise	✓	✓		
Access to Student Resume Pool	✓	✓		
Featured Post on Social Media	✓			
PR Day At Rensselaer Motorsport	✓			



How to Donate

Please feel free to contact us if you have more questions or would like a sponsorship proposal from us with more details and information about funding the team.

Note that all cash donations are eligible for tax write-offs as we are a 501(c)3 organization.

Checks can be made out to RPI Formula SAE and mailed to:

RPI Formula SAE, MANE JEC 5049 110 8th St., Troy, NY 12180

Direct online donations can be made through the Rensselaer Motorsport Impact page:

https://impact.rpi.edu/

If our Impact page isn't up, donations can be made through:

Rensselaer Giving > Other > Formula SAE/Rensselaer Motorsport

Contact Us

Contact Information:

Zayd Farjo, Team Lead
farjoz@rpi.edu, (734) 904-5112

Joey Marchand, Sponsor Relations Lead
marchj5@rpi.edu, (636) 751-2075

Online Presence:

Instagram & Facebook: @rensselaer_mo_torsport

Website: https://rensselaermotorsport.com